Enzyme Discovery Agent

Progress report — what we changed, and how it validated

better discovery | cheaper & cleaner | validated on the benchmark

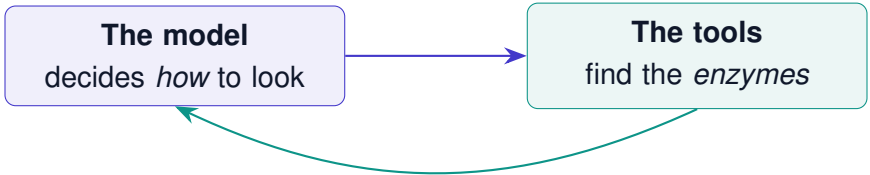
The short version

On our benchmark (find enzymes that join a diacid and a diamine — nylon-6,6 precursors):

- the candidate that was **wrongly ranked #1** — an aromatic amino-acid conjugator — now **drops out of the list entirely**
- a genuine diamine ligase (**PuuA**) is now **#1**, and the exact answer (**DdaG**) moved up to **#3**
- the analogues we were missing (**DesD, AsbA, AsbB, . . .**) are now **all found**
- and it got **cheaper** (~2.18M $\rightarrow$ ~1.43M tokens) with **no slowdown** (~32 min)

Every change ships with the tests passing and the answer still hidden from the agent.

One line of context



The agent discovers enzymes for a target reaction **without ever being told the answer**. The work below makes that discovery **more complete**, its ranking **more honest**, and the whole thing **cheaper**.

What we changed — finding better enzymes

- **Honest substrate matching.** When the target has two substrates, each of the enzyme's natives can now serve only *one* of them — so a candidate can't look good by matching both onto its single best native. *(This is what fixed the wrong #1.)*
- **Keep every enzyme that fits.** We used to keep only a family's best two members; now we keep every member that runs the right direction and has a real substrate to compare. *(This recovered the missing analogues.)*
- **Actually evaluate all of them.** Evaluation is just database lookups, so looking at more is basically free — we now check every candidate, in parallel, instead of stopping at a budget.
- **Read literature more sharply.** Instead of bulk-reading dozens of papers, we read a few and keep only the enzymes they *name* — the one thing literature uniquely gives us (it's how we reach the answer's own gene cluster).

What we changed — cheaper, and cleaner underneath

Cheaper & no slower

- much lighter literature reading (~90% fewer model calls, no stalling on paywalls)
- evaluate candidates in parallel
- **~2.18M → ~1.43M tokens** per run

Cleaner code

- folded nine control gates into one clean pipeline
- removed duplicated scoring / direction logic
- deleted ~730 lines that weren't doing anything

all tests pass **answer still hidden**

None of the cleanup changes behavior — it just makes the pipeline easier to work on.

Did it work? — before vs after

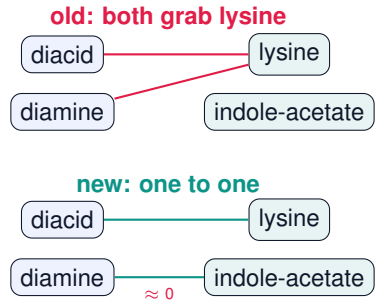
	Before	After
The wrongly-ranked #1 (an aromatic conjugator)	**#1**	**not ranked**
So the #1 candidate is now…	aromatic conjugator	**PuuA (a real diamine ligase)**
DdaG, the exact answer	#4	**#3**
Missing analogues (DesD, AsbA, AsbB, …)	mostly absent	**all recovered**
Tokens per run	~2.18M	**~1.43M**
Wall-clock	~32 min	~32 min

Ground truth is used only to grade the run afterwards — never given to the agent.

Why the old #1 was wrong — and how it's fixed

The old #1 was an enzyme that joins *indole-acetate* to *lysine*. Its lysine is a loose stand-in for one target substrate — and the old code let **both** of our target substrates match onto that same lysine, so it looked far better than it is.

With **one-to-one** matching, the second target substrate is forced onto the enzyme's *other* native — an aromatic indole-acetate, nothing like a diamine — so its score drops to essentially **zero**, and it falls out of the list.



What's still rough — next steps

- **Literature recall.** We still miss some of the field's best papers because we search with the wrong words (and a few are paywalled). Rewriting queries helps, but not reliably yet.
- **Matching a family to its fold.** A few promising families get matched to the wrong structural fold, so they never get a fit score and don't show up on the map — even when one of their members ranks well. Fixing this is the next big win.
- **Speed.** Almost all the time is the model reading long prompts; the wins are shorter prompts and running the final steps in parallel.
- **Ranking.** A couple of scoring measures still don't fully agree — worth unifying, then re-checking ranking quality run over run.

Better answers, honestly ranked, for less.

The wrong #1 is gone, the real answers rose, the missing analogues came back — cheaper, and still with the answer hidden.

honest substrate matching | keep everything that fits | validated